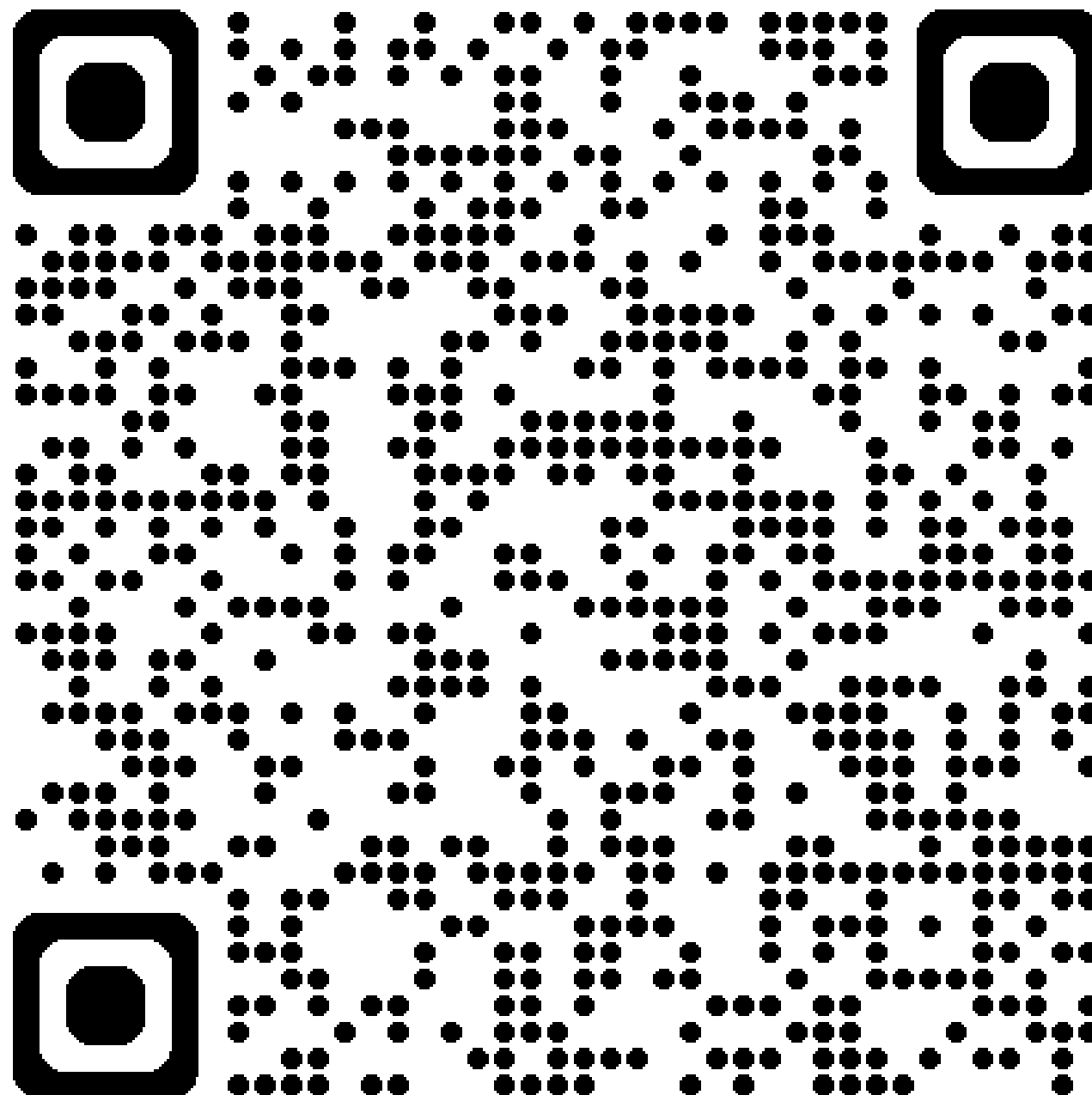
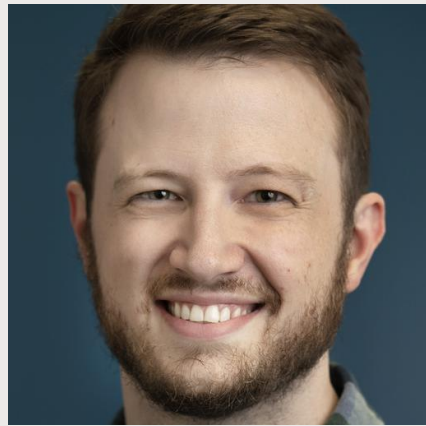


Put An Agent Inside Your App with the GitHub Copilot SDK

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Me



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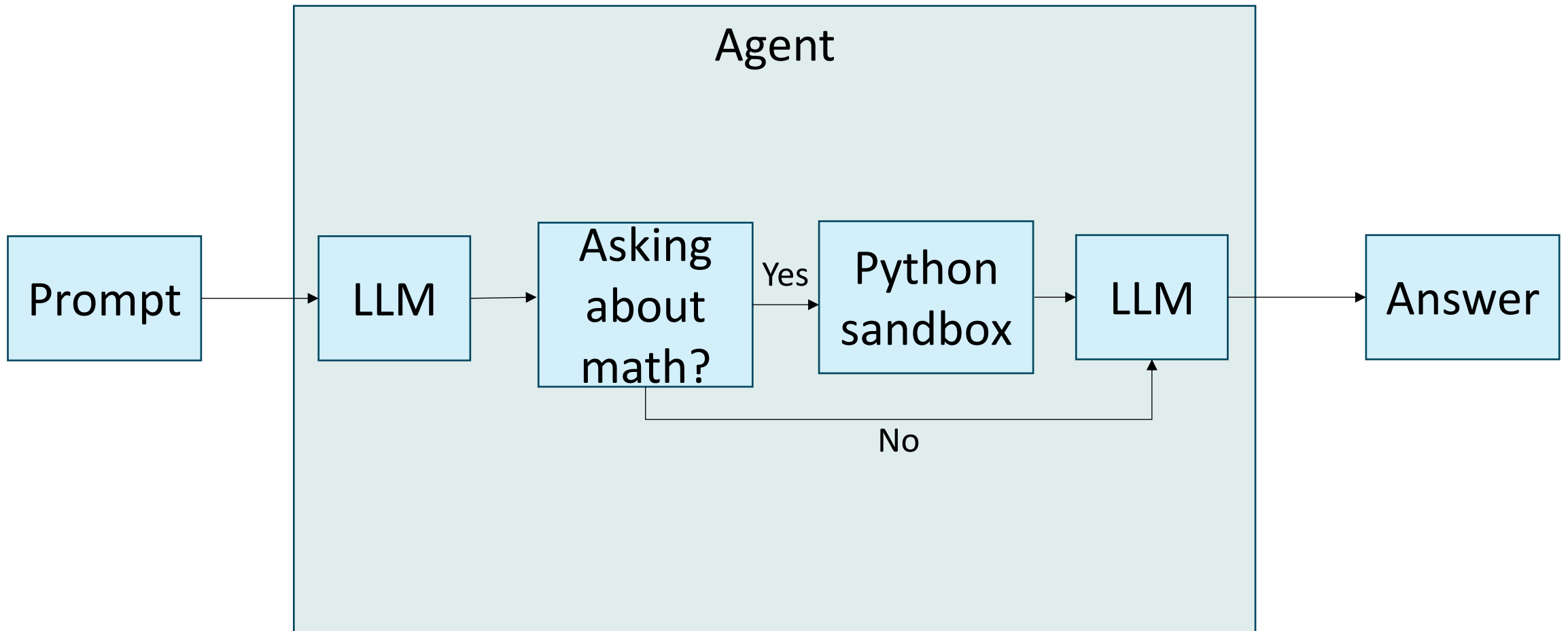
Outline

1. What is an agent?
 2. GitHub Copilot SDK
 3. Why use the SDK?
 4. Coding demo
- 

What is an agent?



What is an agent?

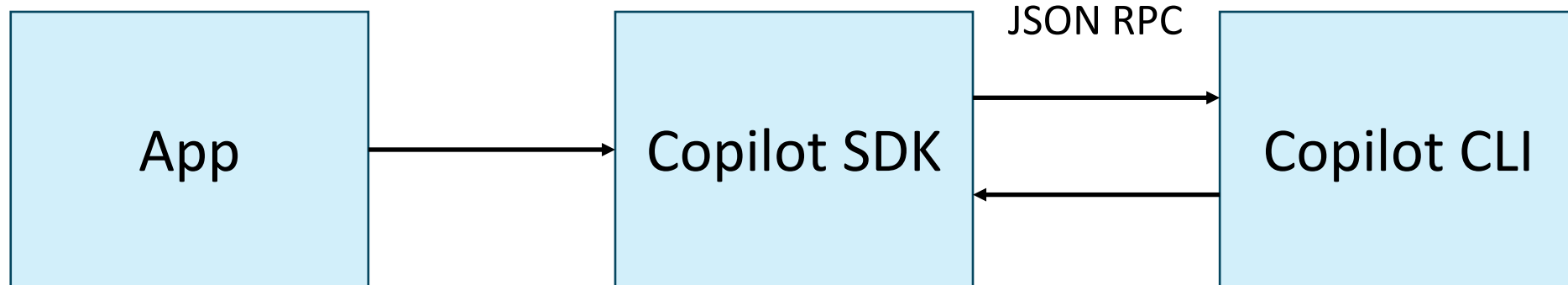


What is an agent?



What is the GitHub Copilot SDK?

- Open-source SDK to call GitHub Copilot from code
 - Official: Typescript, Python, Go, C#
 - Unofficial: Java, Rust, Clojure, C++
- Talks to Copilot CLI through JSON-RPC



GitHub Copilot SDK features

- Need CLI installed on the host machine. Optionally bundle with app
- Everything Copilot normally has in the CLI – agents, skills, MCP, etc.
- Some features are not available by design
 - Export to file, interactive UI, YOLO mode, more
- Supports BYOK – use API key from OpenAI, Azure, Anthropic, etc.
- Auth – signed in CLI user, GitHub OAuth, env vars, BYOK

Tool calling

- Normal code that you register as tools to the LLM

```
[DisplayName("get_weather")]
[Description("Gets the current weather information in the requested city")]
1 reference
WeatherData GetWeather([Description("The name of the city to get the weather for")] string cityName)
{
    return new WeatherData(Temperature: 100, Condition: "It's too dang hot for this time of year");
}
```

```
2 references
record WeatherData(int Temperature, string Condition);
```

```
var session = await client.CreateSessionAsync(new SessionConfig
{
    Model = "gpt-5-mini",
    Tools = [AIFunctionFactory.Create(GetWeather)]
});
```

Wait! Can't I just do that with prompts?

- Yes!
- But...



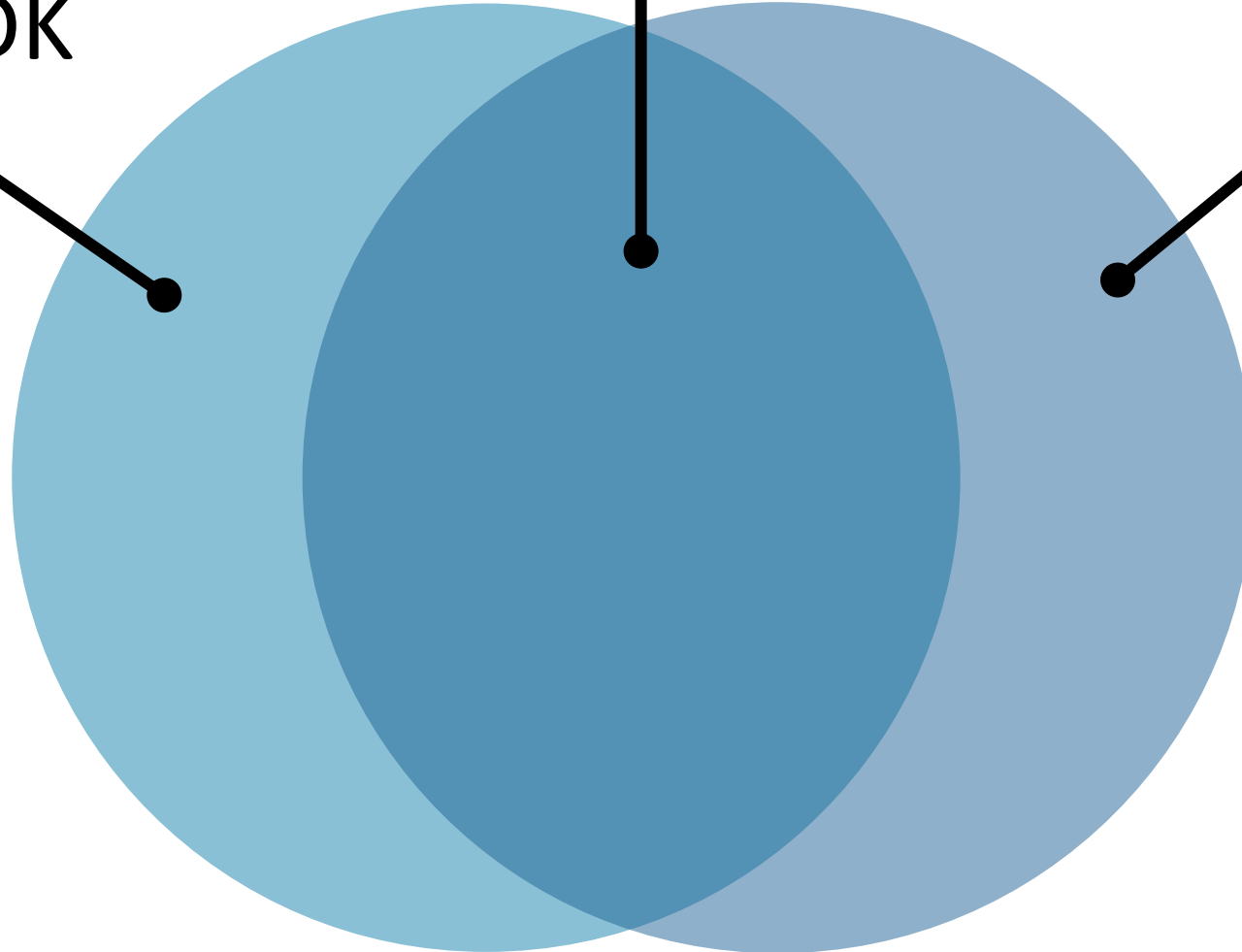
Creative analysis

Normal prompting

Copilot SDK

- App owns the flow
- Put deterministic flow/guardrails around AI analysis

- User owns the flow
- Task is completely exploratory

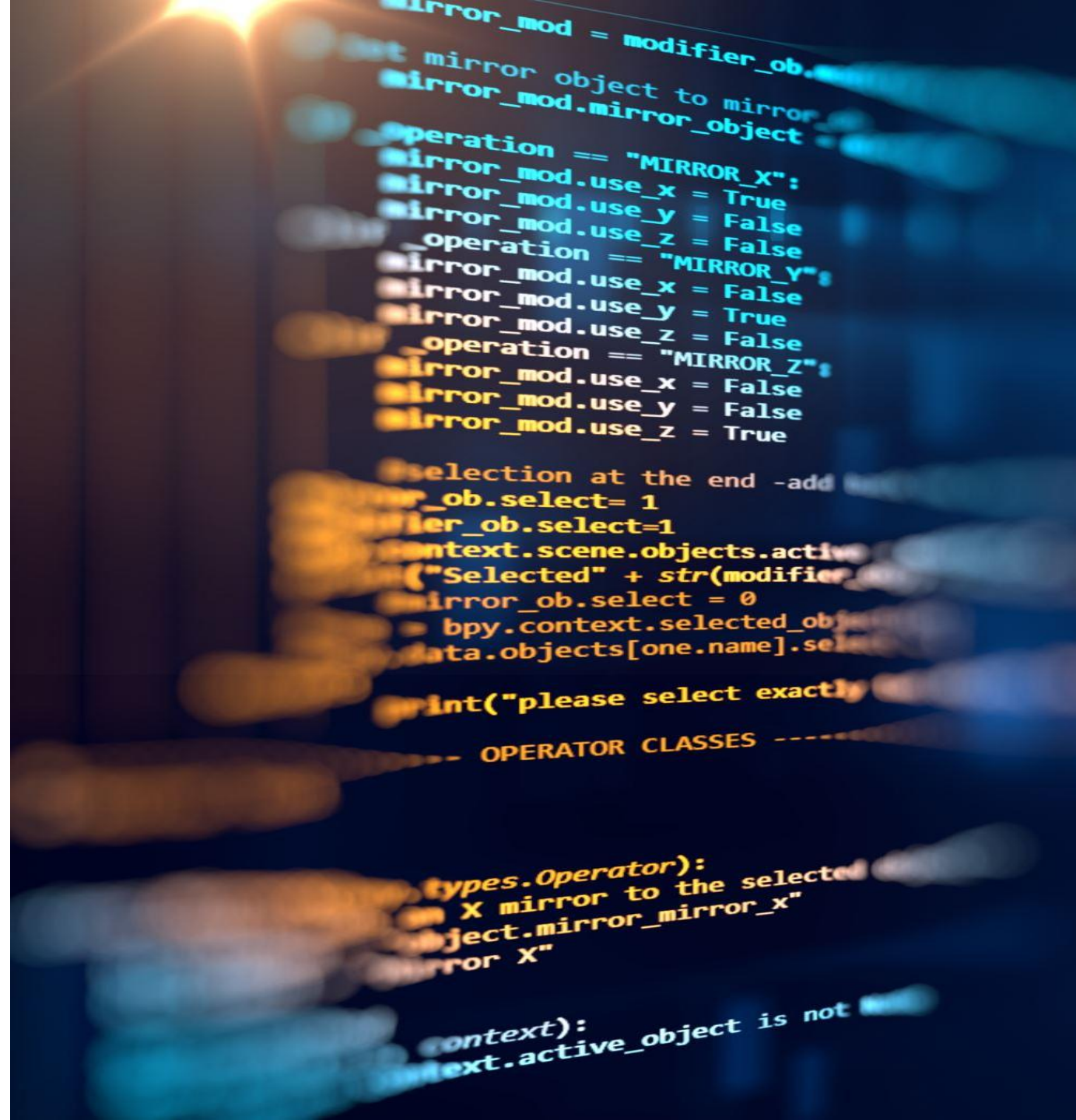


Some examples

Use case	SDK or prompting?	Reason
Triage incoming defects	SDK	Fixed flow; AI pieces things together
Pair programming/code review	Prompting	Context heavy; back-and-forth
In-app product recommendations	SDK	AI embedded in the data flow
Brainstorming product names	Prompting	No tools needed; conversation is the process

Coding Demo

- Requirements
 - Copilot license (free)
 - CLI installed



Phase 1: Planning

-  Locate idea file
-  Generate slug
-  Validate slug regex
-  Copy over scripts, create folder
-  Generate implementation plan + spec files
-  Validate spec count (between 0 and max)

Phase 2: Generation

-  Loop over each spec file
-  Implement the spec fully as code, verify tests pass + builds

Phase 3: Verification

-  Verify all user flows work as expected
-  Generate README

Recap

1. AI agents
2. GitHub Copilot SDK features
3. Why to use the SDK
4. Coding demo

Resources

- <https://github.com/github/copilot-sdk/>
 - Getting Started
 - Cookbook
 - Lots of code samples
 - Open source

Thank you!

